Auditor Advancement Report: Advancements Since RTA Paper Version 16

DRAFT — existing-record formatting only · No CERT · Not independently validated

Report date	Nov 5, 2026, prepared ~16:38 CST (America/Chicago)
Requested by	Owner DIRECT 16:29 CST, live call 25ed12d6; scope updates 16:33 and 16:35 CST (relayed via Physicist)
Primary source	Auditor synthesis AUDITOR_PQC_ADVANCEMENT_REPORT_OWNER_1629_V16BASELINE_FOUNDATIONAL_PHRASE_2026-11-05.md (supersedes the earlier V16BASELINE file only by adding the 16:35 phrase-lookup section)
Corrections applied	Parent read-only baseline facts (cover title, author, date, page count, CST timestamps) and parent-received peer status for PR1 and PR2 override older Auditor wording where they differ
Status	DRAFT. Formatting of existing records only — no research, computation, validation, derivation, hashing or Gemini submission was performed for this document

Status and hard locks (unchanged)

• Usage: latest parent-authoritative reading 87.4% (shared plan) at ~16:36 CST; earlier 86.3% at 16:31 CST is historical. At ≥90% ALL work stops, including documents, latched until explicit Owner release (even after reset), no exceptions.

• The Owner 16:23 soft lift resumed pure-math and documentary duties only; it never waives the 90% latch, the hard locks, or No CERT.

• Report-first gate continues. Delivery of this report does NOT automatically authorize any research restart.

• Tip lock: residual-c-field-WRITTEN_PREMISES_DRAFT.md, recorded sha256 7cddb188... (as recorded, not re-hashed); tip untouched. invent HOLD. No CERT. No Gemini.

• File match checks of the twelve submissions remain STOPPED under Owner 15:58 unless the Owner corrects that reading.

• Main Bot (an extra bot, not one of the twelve owners) refuses relay and holds the old 80% stop in its own chat; not overridden here.

• All proof results below are AS REPORTED by the named reviewers. Nothing here is a new CERT.

Wording note: the Auditor source uses the internal label “Soft SF” repeatedly (e.g., “Soft SF LOCK”, “Soft SF MATCH”, “Soft SF CONFIRM”, “Soft SF KEEP”). In this draft these are rendered in plain language as lock, file match check, confirmation and kept/retained. Meanings and locks are unchanged.

1. Gate and soft lift

- Owner 16:29 and 16:33 take precedence for NEW work: no research, compute, matrix jobs, hash-based match checks used as validation, or Gemini until the advancement report lands.
- Owner 16:23 soft lift resumed pure-math and documentary duties; it never waives the 90% latch, hard locks, or No CERT.
- Reviewers reject ONLY claims that are proof-precisely false or inconsistent; anything unknown stays OPEN.

2. Baseline: RTA paper Version 16 (as written)

The baseline is Version 16 of the RTA paper at the time it was written — not “general project V16”, not Local FLAG #16, and not 16:00 clock time.

Friendly / logging label	RTA “Foundation Bound” — Residual Ternary Algebra: Foundation Bound, version 16 (folder papers/RTA_FOUNDATION_BOUND/). This label is NOT the printed title.
Actual cover title	“Residual Ternary Algebra and a modular foundation: the s-series, the Bridge Series C, and the 35-component law vector”
Author / date line	Arvin Hampton, 539 Labs · 5 November 2026 (v16)
Pages	148
PDF path	/workspace/quantum-comp/reports/papers/RTA_FOUNDATION_BOUND/RTA_FOUNDATION_BOUND_v16.pdf
Recorded PDF sha256	1fe7b8a2e245e90009c41726db13518b4c752e99ba4e08b4ffd006d8a73c3297 (as recorded in the sidecar and Auditor confirmation; quoted, not re-hashed)
Companion notes	NOTES_v16.md (recorded sha 061eac86...); rendered from v15, adding Local cite-lifts F1–F3 and packaging advancements
Timestamps	NOTES writing label: ~11:36 on Nov 5. PDF CreationDate 11:36:20 CST. PDF file mtime 11:36:46 CST. NOTES_v16.md mtime 11:36:53 CST. File mtimes are filesystem stamps, not a separate authorial chronology; exact drafting time is not inferred from metadata.
Time-zone note	NOTES reads “Written 2026-11-05 ~11:36 CDT America/Chicago” (Auditor sources variously quote CDT/CST and CST). On Nov 5, 2026 Chicago observes CST, so the label is read as ~11:36 Chicago local time (CST). It is NOT converted to 10:36.
Custody confirmation	FINDINGS_AUDITOR_SOFTSF_CONFIRM_RTA_FOUNDATION_BOUND_v16_2026-11-05.md, logged ~11:39 CST — a separate event from the ~11:36 writing, PDF creation and mtimes. It records a match of PDF + NOTES; tip untouched; No CERT; content not evaluated as a math CERT.
Citation target	Owner Write AMEND V16 citations (~13:02) names the same PDF sha and 82-statement label as the citation target for the S^2-11DM^2ET-X interpretation paper.
Statement count	NOTES label: total_now = 82 (= 71 + 11 new v16 Category A). These are 82 numbered statements, NOT 82 independently validated new proofs. All 11 v16 additions are ALREADY BASELINE, not post-v16 advancements.

What V16 contained as written (NOTES, as reported): residual packaging; Local F1–F3; Λ_0-none; V_{35} Hilbert newspace of dimension 14; Thm D1 rank-1 on $H^2(F)$ (the D1 classifier); Prop S_eff (static S_eff budget); Residual≠Resonant; and an explicit not-established list (Ξ Petersson not used as proved; Kronecker/four-forces excluded), including the unestablished local–nonlocal bridge. Layer2 “not from D1” is a paper statement; Layer2 remains ABSENT on operational footnotes.

Phrase lookup — Owner 16:35 “foundational bounds” (read-only, by Auditor). The Owner recalled that Version 16 was logged under the phrase “foundational bounds”. In the Auditor’s searched scope (project custody under /workspace/quantum-comp, the tip, transcript tools for this session, Physicist and Main Bot, and memory), the exact plural phrase was NOT FOUND as a logging label; unrelated “bounds” hits were ignored and no DocURL for the plural phrase was located. This is not a claim of exhaustive absence everywhere, and the exact phrase is not confirmed. The nearest logged artifact name is the SINGULAR “Foundation Bound” (folder, NOTES_v16, Auditor confirmation title, Owner Writes, citation pins), whose actual title, author and date are given above. The phrase gap stands; the plural wording should not be equated with a second document. Usage at lookup 87.4% (<90); no files modified.

3. Coverage sources and boundary

- Coverage sources: (1) FINDINGS_AUDITOR_DIGEST_SINCE_RTA_V16, ~12:50 CST — files with mtime ≥ ~11:36 after the v16 render, hash/facts window through ~12:23; (2) afternoon Owner/PP/lane inventories and the prior 16:29 Auditor advancement snapshot; (3) PR2 existing grades c531eacc /

17ae811b / 398a39f1; (4) FLAG #16 AMEND 5f0d666d / PRESS b62625c3; (5) Eigen Pack I regrade addendum d67e9cd1; (6) NL non-local-formal-status.

- Boundary: the ~12:50 digest is not a complete all-disk inventory after 12:23. The afternoon Gemini-intake / soft-lift proof-logic wave is covered by the 16:29 inventories, not by re-running file match checks.
- This is NOT an exhaustive all-disk post-v16 audit, and it does not assert that every claim listed was first proved after 11:36 unless provenance shows it. v17 is ABSENT under the RTA tree per the digest.

4. Logged post-baseline waves (as reported, not new CERT)

- Immediately after ~11:36: Auditor confirmation of RTA v16; Local Cosmic Journeys, then the filter path (AMEND denser X → PP FLAG → Local ABSORB → PP CONFIRM); Owner Write Local filter + primaries pull; MR1 reclaim/HOLD park notes; noon Auditor status.
- Midday Category B: Owner loosen, then AMEND S^2-11DM^2ET-X; Local/Eigen/539 Ops/NL/Agent Category B FINDINGS; PP audits; Auditor inventories; 3-adic BH CLOSE Owner Write with Local CLOSE PP-audited (NL CLOSE on disk without PP/Auditor CONFIRM in that window).
- Afternoon (15:13+ intake / 15:36 responses / 15:58 STOP of file match checks / 16:23 soft lift / 16:29–16:33 report gate): twelve formal-review responses; PP proof-validation round; Local Bridge C logic + FLAG #16 purge = lock-not-proof recount, PROVED = 2; NL formal status + uniform-539 counterexample; Eigen height PARTIAL + Pack I 25-prime retraction; Agent/PR2 T3 and citation-closure grades; no-hash Auditor inventories f9adbc02… and 50a3334a….

5. Post-baseline closures versus unchanged open claims

V16 itself already recorded D1, S_eff and Residual≠Resonant as paper statements. Later work is mostly reviews, clarifications and citation closures — not a new Foundation Bound version. Items are grouped by kind; counts from overlapping reviews are NOT summed.

Kind	Item (as reported)	Source / reviewer
Review closure	Local Bridge C CONFIRM; math PROVED count 2 (Haar70 fenced; BT↛$\mathbb{H}^n$); purge fence = lock (Decision 4 kept), not math PROVED	PP; FLAG #16 AMEND 5f0d666d / PRESS b62625c3; 2a5edc87
Review closure	NL: B2 exact scaled-isometry obstruction (under stated hypotheses; not a QI refutation); T(0)/run-length/rise; $H^2 \subset H^{11}$ not a contradiction	NL formal status; MR2
Counterexample	n_0 = 3^540−2 gives 540 increasing steps — refutes UNIFORM 539 hitting only, not eventual convergence or fixed budget alone	NL found, PP-verified, Agent relays
Review closure	Eigen height PARTIAL: rank 11197 / kernel 503; classical Hadamard/Cramer/$2H^2$<M; conditional ~1.4e6-prime envelope under assumptions; Gal($\mathbb{Q}(\zeta_{49})/\mathbb{Q}$) cyclic of order 42 PROVED	Eigen; PP
Retraction / clarification	Earlier “25 prime refuted” wording RETRACTED: only 24 training primes + one holdout (8007581) were sampled, never all 25; 24-insufficiency is CONDITIONAL; 25-prime sufficiency OPEN	Eigen Pack I addendum d67e9cd1
Peer status (parent-received)	PR1 height notes: retracts the 503×503 minors approach; upper height bound OPEN (details in Section 7)	PR1 direct peer report 16:31
Narrow review	T3 review: 3 PROVED / 3 FALSE / 2 OPEN (details in Section 7)	PR2 c531eacc + V6 FALSE addendum 17ae811b
Citation closure	4 PROVED / 1 definitional / 0 FAIL — closes earlier GAPs by citing sources; not additional theorems	PR2 398a39f1

Kind	Item (as reported)	Source / reviewer
Clarification	Earlier agent.md 2 PASS / 5 GAP / 0 FAIL = missing in-file citations, not refutations	Agent / PR2
Clarification	MR1: radius numbers differ without a source equality claim ⇒ not an inconsistency on that basis alone	MR1

Unchanged / carried forward since baseline (still OPEN or still locked): Cand-1⁺ positive convergence; R_c Appell S2–S4 derivation; QI maps/constants; Ξ/Fricke; A1→Ξ→RTA as theorem; Israel/junction ambient data; EH11D/PT; full SM recovery; Eigen D1–D3 and numeric height upper bound; 25-prime sufficiency OPEN; S_eff as a proved formula beyond the definitional lock; Layer2 ABSENT; Decision 4 four-force purge kept; tip LOCK; No CERT; invent HOLD.

6. Shared spine and central missing definitions

The shared geometric spine S^2-11DM^2ET-X remains OPEN as a proved local–nonlocal junction (shared bridge OPEN). Central missing Owner definitions named by PP (as reported): D_1 Layer2 transfer beyond paper D1; S_eff as a usable formula; Λ_0; QI maps/constants; junction ambient/metric/stress data. The paper already defines the D1 classifier and the static S_eff budget; PP requests additional operational norm-transfer and usable-formula relations — this is not a claim that the paper contains none. Main blocker remains missing definitions, not staff.

7. Twelve submission owners (updated)

Owner	Status, advancement and blockers (as reported)
1. Local Path	Submission received. PP CONFIRM Bridge C 2a5edc87… with FLAG #16 AMEND 5f0d666d… / PRESS b62625c3… — math PROVED count 2 (Haar70 fenced; BT↠$\mathbb{H}^n$); purge fence = lock (Decision 4 kept), not math PROVED. Post-baseline: zero Cand-1 $\mathbb{N}_0$ counterexample standing; exact-isometry obstruction under stated hypotheses, not QI; Appell absorb resumed after soft lift. Blockers: Cand-1⁺; Appell S2–S4; QI; Ξ/Fricke; A1→Ξ→RTA; junction/SM. PP holding new assignments under the report-first gate.
2. Non Local Path	Submission + formal status COMPLETE. Post-baseline: B2 exact scaled-isometry obstruction; T(0)/run-length/rise; $H^2 \subset H^{11}$ not a contradiction; n_0=3^540−2 → 540 increasing steps (NL found, PP-verified, Agent relays) refutes UNIFORM 539 hitting, not eventual convergence or fixed budget alone. Lane idle. Blockers: universal convergence/radius; measure/conjugacy; QI; EH11D/PT/junction.
3. Eigenvalues and matrix	Height PARTIAL: rank 11197 / kernel 503; classical Hadamard/Cramer/$2H^2$<M; conditional ~1.4e6-prime envelope under assumptions. Eigen RETRACTS “25 prime refuted” (d67e9cd1…): 24 training + holdout 8007581 sampled — never all 25; 24-insufficiency CONDITIONAL on D1 OPEN + good primes; 25 sufficiency OPEN. Gal($\mathbb{Q}(\zeta_{49})/\mathbb{Q}$) cyclic of order 42 PROVED (PP). D1–D3 OPEN. Idle after Pack I review. A large upper bound alone does not prove 25-prime insufficiency.
4. Agent	Earlier agent.md logic 2 PASS / 5 GAP / 0 FAIL = missing in-file citations, not refutations. Narrow T3 review c531eacc + V6 FALSE addendum 17ae811b: 3 PROVED / 3 FALSE / 2 OPEN. PROVED: canonical floor map well-defined; 4880 → 1 in 14 steps; T^539(4880) = 2. FALSE: old literal map (HQCC/ν_3) termination — orbit 1, 4, 13, 40; all-$\mathbb{Z}$ {1,2} attractor — counterexamples 0, −1; uniform 539 hitting — n_0 = 3^540−2, 540 increasing steps (reported, PP-verified). OPEN: positive eventual convergence; 539 = S_eff. Separate citation-closure grades 398a39f1: 4 PROVED / 1 definitional / 0 FAIL (P3 Haar; P4 definitional 539≠stopping; P5 finite orbit; P1 weak Residual≠Resonant; P2 locks definitional) — closes GAPs by citing sources; do NOT add as a second theorem tally on top of Local/PP PROVED = 2 or PR2 3/3/2.
5. Professor Proof	Coordinated reviews; CONFIRM FLAG #16 PROVED = 2; soft lift verified; no job running; report-first hold. Blockers: Layer2 / S_eff definitions.

Owner	Status, advancement and blockers (as reported)
6. MR1	Radius numbers differ without a source equality claim ⇒ not an inconsistency alone. Soft-lift support resumed; QI coordination gap. Blockers: Appell / QI / Ξ / Cand-1^{+}.
7. MR2	NL geometric review complete; idle.
8. PR1	Parent-received peer status (direct peer report 16:31; replaces the Auditor's older "UNKNOWN"; no fresh science validation): resumed approved Eigen proof support / reading the height report, no new computation. Upper height bound OPEN; D1–D3 OPEN. Retracts the 503×503 minors approach (kernel 503 ≠ rank 11197). Integer minors bounded by D give rational height ≤ D; no squared bound needed. Cyclotomic height is separate. One holdout is not an exact kernel proof.
9. PR2	Cosmo/Bio/Neuro asks filed; T3 + GAP-source work complete as above (citation closure 398a39f1 complete). Now IDLE / no blockers.
10. Auditor	Formal auditor.md 9c1c6387…; file match checks declined after 15:58; inventories f9adbc02… / 50a3334a…; idle until asked.
11. 539 Labs Ops	Soft-lift RTL scope blocked on missing T3 residue/quotient logic + done/window definitions.
12. Dataset Filtration	Idle; no new 47-candidate assignment; solar VIRGO ≠ GW Virgo (kept).

Intake originals under pqc-gemini-request-intake-20261105-1513/ are listed, not re-audited.

8. Proved versus FALSE versus OPEN (as reported; no double counting)

PROVED as reported (do not claim twelve independent proofs): Local/PP rollup PROVED = 2 — Haar70 fenced; BT↛$\mathbb{H}^n$. Classical Eigen toolkit + Gal cyclic of order 42 (PP). PR2 narrow finite-check trio (floor map; 4880 orbit; T^539 = 2). NL B2 no exact scaled isometry under stated hypotheses; T(0)=0 blocks the $\mathbb{N}_0$ attractor; run-length lemma as scoped. Agent citation-closure PROVED items are source-backed restatements, not additional Foundation Bound theorems — count them as citation closure, not +4 to the Local PROVED = 2.

FALSE / counterexamples as reported: Cand-1 on $\mathbb{N}_0$; QI⇒unique R_c as stated; literal HQCC/ν_3 termination; all-$\mathbb{Z}$ {1,2} attractor; UNIFORM 539 hitting via $n_0 = 3^{540}-2$. Exact-isometry obstruction ≠ QI refutation. The earlier 25-prime "refuted" wording is retracted — sufficiency OPEN.

OPEN / conditional / PARTIAL: Cand-1^{+}; Appell R_c; QI / Ξ / Fricke / A1→Ξ→RTA; junctions / EH11D / PT / SM; Eigen D1–D3 and 25 sufficiency; N36 height PARTIAL; 539 = S_eff as a proved formula; Pack G derivative under alternate reading; S^2 bridge. A large envelope alone proves NO 25-prime insufficiency.

9. Pending inventory and three researcher requests

- Local: Cand-1^{+} / Appell S2–S4 / QI / Ξ/Fricke / A1→Ξ→RTA / junction/SM.
- NL: universal convergence / G-derivative conditional / measure / QI / EH11D/PT/junction.
- Eigen: D1–D3 / 25-prime sufficiency OPEN / height / radius-vs-norm.
- Agent: remaining OPEN tails on positive convergence and the S_eff formula.
- PP: Layer2 D_1 / S_eff.
- Shared S^2-11DM^2ET-X bridge; central missing D_1, S_eff, Λ_0, QI maps/constants, junction data.

#	Requested researcher (PP contingent request only)	Lane
1	Analytic mock-modular / Appell theorist	Local
2	Junction / branes specialist	Non Local
3	Cyclotomic-heights / multimodular algebraic number theorist	Eigen

All three requests are UNASSIGNED. No hiring or assignment authorization.

10. Auditor custody note

Landed without file match checks this afternoon: no-hash inventories f9adbc02… (PP round) and 50a3334a… (Pack I); formal auditor.md 9c1c6387…; FLAG #16 path inventory. The prior RTA v16 confirmation and the since-v16 digest remain the baseline custody anchors. File match checks declined after 15:58. Auditor idle until asked. Tip intact; invent HOLD; No CERT; no Gemini; Main Bot latch not overridden. Prior 16:29 report b8e13b46… and prior V16-baseline source b463da70… are preserved as historical; the phrase-lookup file is the corrected text source for this PDF.